

A Formally Verified Divisibility Theorem for A211420

Self-contained mathematical proof and kernel-checked Lean 4 formalization

21 July 2026

Contribution statement

The ordinary mathematical argument was generated with OpenAI GPT-5.6 Pro. OpenAI Codex produced and kernel-checked the Lean 4 formalization and prepared the repository. Danny Cabezas initiated, curated, and published the project.

1. The theorem

For natural numbers n , r , and k , define

$$\begin{aligned} \text{ANum}(n) &= (8^n)! \cdot n! \\ \text{ADen}(n) &= (4^n)! \cdot (3^n)! \cdot (2^n)! \\ D(k, r, n) &= \text{product over } i = 1, \dots, r \text{ of } (k^n + i) \\ L(r) &= \text{lcm}(1, 2, \dots, 8^r - 1) = \text{Nat.lcmUpto } (8^r - 1) \end{aligned}$$

Strengthened A211420 divisibility

For every $n \geq 0$, every $r \geq 1$, and every k in $\{1, 2, 3\}$, $\text{ADen}(n)$ divides $\text{ANum}(n)$ and

$D(k, r, n) \cdot \text{ADen}(n)$ divides $L(r)^r \cdot \text{ANum}(n)$.

Thus the exact integer quotient $a(n) = \text{ANum}(n) / \text{ADen}(n)$ satisfies $D(k, r, n)$ divides $L(r)^r \cdot a(n)$, and $L(r)^r > 0$.

The cross-multiplied assertion is primary. The quotient is introduced only after integrality, so natural-number division cannot conceal truncation.

2. Landau step function and integrality

For a prime p , initially interpret v_p as the valuation on positive rational numbers. Define

$$\Delta(x) = \text{floor}(8x) + \text{floor}(x) - \text{floor}(4x) - \text{floor}(3x) - \text{floor}(2x).$$

Since $8+1-4-3-2=0$, Δ is one-periodic. Its complete breakpoint list on $[0,1)$ is

$$0, \frac{1}{8}, \frac{1}{4}, \frac{1}{3}, \frac{3}{8}, \frac{1}{2}, \frac{5}{8}, \frac{2}{3}, \frac{3}{4}, \frac{7}{8}, 1.$$

Direct evaluation on the intervening half-open intervals gives

Exact support of Δ

$\Delta(x)=1$ exactly when the fractional part of x belongs to

$$[\frac{1}{8}, \frac{1}{3}) \cup [\frac{3}{8}, \frac{1}{2}) \cup [\frac{5}{8}, \frac{2}{3}) \cup [\frac{7}{8}, 1).$$

It is zero elsewhere. In particular, $\Delta(x) \geq 0$.

Legendre's formula yields the finite valuation identity

$$v_p(\text{ANum}(n)/\text{ADen}(n)) = \sum_{j \geq 1} \Delta(n/p^j).$$

All summands vanish after $p^j > 8n$. Every prime valuation of the positive rational quotient is nonnegative. Therefore $\text{ADen}(n)$ divides $\text{ANum}(n)$, and the factorial ratio $a(n)$ is a positive integer.

This establishes exact quotient recovery before any integer divisibility statement involving $a(n)$ is used.

3. Prime powers in the consecutive product

For a prime power $q=p^j$, put

$$N_q = \text{number of } i \text{ in } \{1, \dots, r\} \text{ such that } q \text{ divides } k \cdot n + i.$$

Then $v_p(D(k, r, n)) = \sum_{j \geq 1} N_{(p^j)}$. The sum is finite.

Assume first that $q \geq 8 \cdot r$. Since $q > r$, at most one of the r consecutive terms is divisible by q . If q divides $k \cdot n + i$, write $k \cdot n + i = m \cdot q$. Then

$$n/q = m/k - i/(k \cdot q), \quad 0 < i/(k \cdot q) \leq 1/(8 \cdot k).$$

Sorting m modulo k places the fractional part of n/q in the following intervals:

k	possible intervals
1	$[7/8, 1)$
2	$[15/16, 1)$ or $[7/16, 1/2)$
3	$[23/24, 1)$, $[7/24, 1/3)$, or $[5/8, 2/3)$

Each interval is contained in the support of Δ . The included left endpoints cover the delicate equality case $q=8 \cdot r$ and $i=r$. Hence

$$N_q \leq \Delta(n/q) \quad \text{whenever } q \geq 8 \cdot r.$$

For a small prime power $q=p^j < 8 \cdot r$, simply use $N_q \leq r$. The exact least common multiple satisfies

$$v_p(L(r)) = \text{number of } j \geq 1 \text{ such that } p^j < 8 \cdot r.$$

4. Valuation comparison and conclusion

For every prime p , split the valuation sum at $8 \cdot r$:

$$\begin{aligned} v_p(D(k, r, n)) &= \sum_{p^j < 8 \cdot r} N(p^j) + \sum_{p^j \geq 8 \cdot r} N(p^j) \\ &\leq r \cdot v_p(L(r)) + \sum_{p^j \geq 8 \cdot r} \Delta(n/p^j) \\ &\leq r \cdot v_p(L(r)) + \sum_{j \geq 1} \Delta(n/p^j) \\ &= v_p(L(r)^r \cdot a(n)). \end{aligned}$$

The second inequality uses $\Delta \geq 0$ for the omitted small powers. Prime-by-prime comparison proves $D(k, r, n)$ divides $L(r)^r \cdot a(n)$. The least common multiple is nonzero, so its r -th power is positive.

5. Lean verification

The checked development uses natural-number factorization and mathlib's exact factorial and lcm valuation APIs. Its public declarations include:

```
A211420.integral_cross
A211420.strengthened_cross
A211420.constant_pos
A211420.ADen_mul_a
A211420.strengthened
```

The primary declaration is cross-multiplied. The quotient corollary follows only after exact cancellation. The project is pinned to Lean 4.30.0 (d024af099ca4bf2c86f649261ebf59565dc8c622) and mathlib v4.30.0 (c5ea00351c28e24afc9f0f84379aa41082b1188f).

Run `./scripts/verify.sh` from the repository root. The script builds the project and rejects forbidden project proof escapes.

6. Verification status and scope

Check	Result
Lean kernel build	PASS - 8,476 jobs
Forbidden-token negative control	PASS - the injected placeholder was rejected
Public proof source SHA-256	2630d746196a362d5d7cc438e8ba9556f716ece9ba61814c33a94df1a51bef9d

The displayed theorem is kernel-checked. Finite computations were diagnostic only and are not used as proof. The two adversarial reports in the repository are AI-generated checks, not human peer review.

No claim is made that the constant is minimal, that the proof has historical priority, or that additional variants in research drafts have been formalized. The supplied searches found no prior source for this exact $L(r)^r$ strengthening, but a bounded search cannot certify originality.

Background references

E. Landau, *Sur les conditions de divisibilite d'un produit de factorielles par un autre*, Nouvelles annales de mathematiques 19 (1900), 344-362.

J. W. Bober, *Factorial ratios, hypergeometric series, and a family of step functions*, Journal of the London Mathematical Society 79 (2009), 422-444.

F. Rodriguez-Villegas, *Integral ratios of factorials and algebraic hypergeometric functions*, arXiv:math/0701362 (2007).